UNIVERSITAT POLITÈCNICA DE CATALUNYA
BARCELONATECH
Departament de Mecànica de Fluids

# FORMULARI TURBOMÀQUINES

## Circuit hidràulic

$$\frac{p_1}{\rho g}+\frac{c_1^2}{2g}+z_1 \pm H_{generadora/turbina}=\frac{p_2}{\rho g}+\frac{c_2^2}{2g}+z_2+h_f+h_s$$

$$h_f=\frac{8fLQ^2}{\pi^2 D^5 g}=f\frac{L}{D}\frac{c^2}{2g}$$

$$h_s=K_s\frac{c^2}{2g}=\frac{8K_sQ^2}{\pi^2 D^4 g}=K_s'Q^2$$

$$H_o+h_{tot}=H_{generadora}$$

$$H_{brut}-h_{tot}=H_{turbina}=H_{net}$$

$$\rho_{Aigua}=1000\frac{Kg}{m^3};\rho_{Aire}=1.2\frac{Kg}{m^3}$$

## Màquina generadora (bomba o ventilador)

$$H_{gen.}=\frac{C_{out}^2-C_{in}^2}{2g}+(Z_{out}-Z_{in})+\frac{(P_{out}-P_{in})}{\rho g};H_{gen}: H_{bomba}, H_{ventilador}$$

$$P_{mec}=P_{elect}\cdot\eta_{motor}$$

$$P_{hidr}=P_{mec}\cdot\eta_{gen}$$

$$P_{hidr}=\rho\cdot g\cdot Q\cdot H_{gen}$$

$$\eta_{gen}=\frac{P_{hidr}}{P_{mec}}=\frac{\rho g Q H_{gen}}{\omega T}=\eta_h\eta_v\eta_m$$

$$\eta_h=\frac{H}{H_{th}}$$

$$\eta_v=\frac{Q}{Q_{th}}$$

## Màquina receptora (turbina)

$$H_{turbina}=H_n=\frac{C_{in}^2-C_{out}^2}{2g}+(Z_{in}-Z_{out})+\frac{(P_{in}-P_{out})}{\rho g}$$

$$P_{hidr}=\rho\cdot g\cdot Q\cdot H_n$$

$$P_{mec}=P_{hidr}\cdot\eta_{turb.}$$

$$P_{elect}=P_{mec}\cdot\eta_{alternador}$$

$$\eta_{turbina}=\frac{P_{mec}}{P_{hidr}}=\frac{\omega T}{\rho g Q H_n}=\eta_h\eta_v\eta_m$$

$$\eta_h=\frac{H}{H_n}$$

$$\eta_v=\frac{Q}{Q_{th}}$$

## Cinemàtica

$$\vec{C}=\vec{U}+\vec{W}$$

$$i=\beta_{1alep}-\beta_{1fluid}$$

$$d=\beta_{2alep}-\beta_{2fluid}$$

$$E_{th_euler}=gH_{th_euler}=U_2\cdot C_{2u}-U_1\cdot C_{1u}$$

$$E_{th_euler}=\left(\frac{C_2^2-C_1^2}{2}\right)+\left(\frac{U_2^2-U_1^2}{2}\right)+\left(\frac{W_1^2-W_2^2}{2}\right)$$

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# FORMULARI MÀQUINES HIDRÀULIQUES

$$E_{th_euler} = U_2 \cdot C_{2m} cotan\ \alpha_{2f} - U_1 \cdot C_{1m} cotan\ \alpha_{1f}$$

$$H_{th_euler} = \frac{U_2 \cdot \left(U_2 - C_{2m}\ cotan\beta_{2f}\right)}{g} = \frac{(\omega R_2)^2}{g} - \frac{\omega cotan\beta_{2f}}{2\pi b_2 g} Q$$

## Nombres adimensionals (bombes)

$$\varphi = \frac{Q}{\Omega D^3} \qquad \Psi = \frac{E}{\Omega^2 D^2} \qquad \lambda = \frac{P_M}{\rho \Omega^3 D^5} \qquad Re = \frac{\rho \Omega D^2}{\mu} \qquad \delta = \frac{\epsilon}{D}$$

$$\varphi' = \frac{Q}{nD^3} \qquad \Psi' = \frac{H}{n^2 D^2} \qquad \lambda' = \frac{P_M}{\rho n^3 D^5}$$

## Nombres adimensionals (turbines)

$$Q_{ED} = \frac{Q}{E^{0.5} D^2} \qquad n_{ED} = \frac{\Omega D}{E^{0.5}} \qquad P_{ED} = \frac{P_M}{\rho E^{1.5} D^2} \qquad n_s = \frac{n Q^{1/2}}{E^{3/4}}$$

## Semblança

$$\frac{H_2}{H_1} = \left(\frac{n_2}{n_1}\right)^2 \left(\frac{D_2}{D_1}\right)^2 \qquad \frac{Q_2}{Q_1} = \left(\frac{n_2}{n_1}\right)\left(\frac{D_2}{D_1}\right)^3$$

$$\frac{P_{M2}}{P_{M1}} = \left(\frac{\rho_2}{\rho_1}\right)\left(\frac{n_2}{n_1}\right)^3 \left(\frac{D_2}{D_1}\right)^5 \qquad \frac{1-\eta_p}{1-\eta_m} = \left(\frac{D_m}{D_p}\right)^{0.25}$$

## Cavitació

$$NPSH_{d,1} = \frac{P_{0abs}}{\gamma} - h_{tot,0-1} + H_{gen,0-1} + (z_0 - z_1) - \frac{P_{vapor}}{\gamma} = \frac{P_{1abs}}{\gamma} + \frac{C_1^2}{2g} - \frac{P_{vapor}}{\gamma}$$

## Aerogenerador

$$P_{mec} = C_p \frac{1}{2} \rho A c_1^3$$

$$C_{p-Betz} = 0.593$$

$$\lambda = \frac{2\pi r n}{60 c_1}$$

β, “pitch”

